



PHY 110 MTE-1

B.tech CSE (Lovely Professional University)



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Course Code: PHY110
Course Title: ENGINEERING PHYSICS

Max Marks: 30

Time Allowed: 01:30hrs.

Read the following instructions carefully before attempting the question paper.

1. Match the Paper Code shaded on the OMR Sheet with the Paper code mentioned on the question paper and ensure that both are the same.
2. This question paper contains 30 questions of 1 mark each. 0.25 marks will be deducted for each wrong answer.
3. All questions are compulsory.
4. Do not write or mark anything on the question paper and/or on rough sheet(s) which could be helpful to any student in copying, except your registration number on the designated space.
5. Submit the question paper and the rough sheet(s) along with the OMR sheet to the invigilator before leaving the examination hall.

- Q(1) What is the fundamental process by which lasers generate light?
(a) Absorption of light (b) Spontaneous emission of light
(c) Stimulated emission of light (d) Refraction of light CO1,L2
- Q(2) Which of the following is NOT one of the three types of energy levels in an atom?
(a) Ground state (b) Excited state (c) Metastable state (d) Intermediate state CO1,L2
- Q(3) Which of the following is a way in which matter can interact with electromagnetic radiation?
(a) Absorption (b) Refraction (c) Reflection (d) Interference CO1,L2
- Q(4) What happens to an atom when it absorbs a photon of light?
(a) It moves to a lower energy level (b) It moves to a higher energy level
(c) It emits a photon of light (d) Nothing happens CO1,L2
- Q(5) Which of the following is a type of emission of light that occurs naturally and randomly?
(a) Absorption (b) Spontaneous emission (c) Stimulated emission (d) Refraction CO1,L2
- Q(6) Which of the following is a type of emission of light that occurs as a result of the interaction between an incoming photon and an excited atom?
(a) Absorption (b) Spontaneous emission (c) Stimulated emission (d) Refraction CO1,L2
- Q(7) What is population inversion in a laser?
(a) When more atoms are in the ground state than the excited state
(b) When more atoms are in the excited state than the ground state
(c) When there are equal numbers of atoms in the ground and excited states
(d) When no atoms are in the ground or excited state CO1,L2
- Q(8) What is the lasing medium in a Nd:YAG laser?
(a) Neon (b) neodymium (c) aluminium (d) Silicon CO1,L2
- Q(9) What is the typical wavelength of a He-Ne laser?
(a) 632.8 nm (b) 1064 nm (c) 1550 nm (d) 980 nm CO1,L2
- Q(10) What is the excitation mechanism in a Nd:YAG laser?
(a) Electric discharge (b) Optical pumping
(c) Electron beam excitation (d) Chemical reaction CO1,L2
- Q(11) In an optical fibre, relation between refractive index of core (n_1) and cladding (n_2) is
(a) $n_1 = n_2$ (b) $n_1 > n_2$ (c) $n_1 < n_2$ (d) $n_1 \leq n_2$ CO1,L2
- Q(12) In an optical fibre, light is guided by means of
(a) Diffraction (b) Interference (c) Total internal reflection (d) polarization CO1,L2
- Q(13) Core of an optical fibre is made of transparent rod of
(a) clear plastic or glass (b) aluminium (c) copper (d) none of these CO1,L2
- Q(14) Angle of acceptance is maximum for a fiber, when
(a) the critical angle is minimum (b) the critical angle is zero
(c) the critical angle is maximum (d) the critical angle is negative CO1,L2

Q(15) Total internal reflection takes place when light travels from

- (a) rarer to denser medium (b) denser to rarer medium
(c) air to glass (d) glass to air

CO1,L2

Q(16) Numerical aperture of an optical fibre is equal to

- (a) $n_1 \sqrt{2\Delta}$ (b) $\frac{n_1}{n_2} \sqrt{2\Delta}$ (c) $\frac{n_1}{\sqrt{2\Delta}}$ (d) $\frac{n_2}{\sqrt{2\Delta}}$

CO1,L2

Q(17) The numerical aperture of an optical fibre depends on

- (a) core refractive index (b) critical angle (c) both (a) and (b) (d) none of these

CO1,L2

Q(18) In an optical fibre, the propagation angle of light must be equal to or less than

- (a) acceptance angle (b) incident angle (c) critical angle (d) refraction

CO1,L2

Q(19) In an optical fibre, dispersion means

- (a) pulse distortion (b) pulse narrowing (c) pulse rise-time (d) pulse broadening

CO1,L2

Q(20) In Graded Index optical fibres (multimode)

- (a) the refractive index of the core has a constant value
(b) the refractive index of the core is equal to that of the cladding
(c) the refractive index of the core decreases continuously between the axis and the cladding
(d) none of these

CO1,L2

Q(21) Solenoidal condition is related to

- (a) grad of vector is zero (b) divergence of vector is zero
(c) curl of vector is zero (d) all of these

CO1,L2

Q(22) Gradient is a

- (a) Vector (b) Scalar (c) Both a and b (d) None of these

CO1,L2

Q(23) Divergence of any vector is

- (a) Vector (b) Scalar (c) Both A and B (d) None of these

CO1,L2

Q(24) The conservative charge of any vector A is related to

- (a) $\text{Curl} A = 0$ (b) $\text{Div} A = 0$ (c) $\text{Grad} A = 0$ (d) None of these

CO1,L2

Q(25) Maxwell's second equation is related to

- (a) $\int \mathbf{E} \cdot d\mathbf{l} = 0$ (b) $\int \mathbf{B} \cdot d\mathbf{s} = 0$ (c) $\int \mathbf{E} \cdot d\mathbf{v} = 0$ (d) None of these

CO1,L2

Q(26) What is the relationship between magnetic field intensity, magnetic flux density, and magnetic permeability?

- (a) $\mathbf{B} = \mu_0 \mathbf{H}$ (b) $\mathbf{H} = \mu_0 \mathbf{B}$ (c) $\mathbf{H} = \mathbf{B} / \mu_0$ (d) None of these

CO1,L2

Q(27) How is the electric field is related to charge carrying conductor, where E is electric field, V is potential difference and L is the length of the conductor?

- (a) $V = E/L$ (b) $E = VL$ (c) $V = EL$ (d) $E = V/L$

CO1,L2

Q(28) Poisson equation is related to

- (a) $\nabla^2 V = -\rho$ (b) $\nabla^2 V = -\frac{\rho}{\epsilon_0}$ (c) $\nabla^2 V = \rho \epsilon_0$ (d) $\nabla^2 V = -\rho \epsilon_0$

CO1,L2

Q(29) Maxwell's first equation is related to

- (a) Electrostatic Gauss law (b) Magnetostatic Gauss law
(c) Faraday's law (d) Ampere's law

CO1,L2

Q(30) Maxwell's third equation is related to

- (a) Electrostatic Gauss law (b) Magnetostatic Gauss law
(c) Faraday's law (d) Ampere's law

CO1,L2

--End of Question paper--